Medical Microbiology

Third stage- College of Dentistry

Lab. 11

Bacillus

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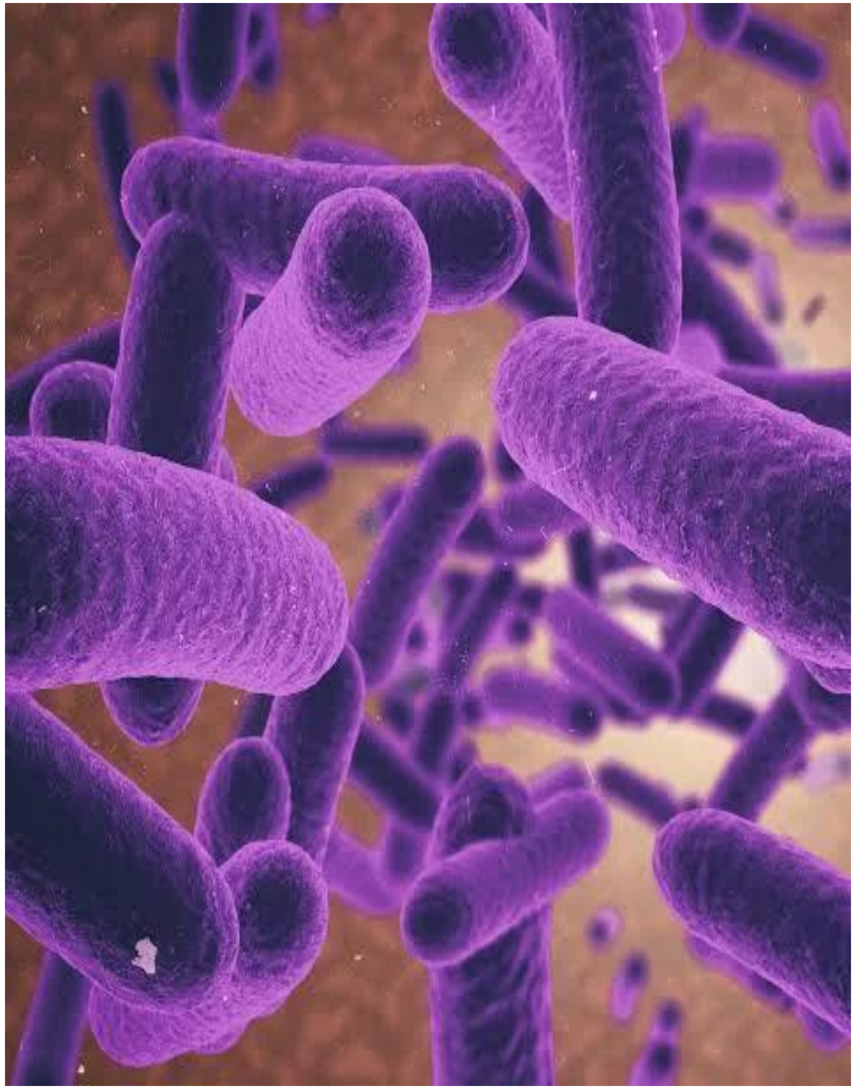
Learning Objectives:

After completing this lab, you should be able to:

1-Describe the microscopic appearance of Bacillus .

2-Explain the laboratory methods used to detect Bacillus.

3- Explain types of biochemical test for detect of Bacillus.



General characteristics of Bacillus

Morphology: They are rod-shaped bacteria.

Gram Stain: Gram-positive in young cultures, though they may become Gram-variable or Gram-negative in older cultures.

Respiration: Aerobic or Facultative Anaerobic.

Habitat: Widely distributed in soil and the environment.

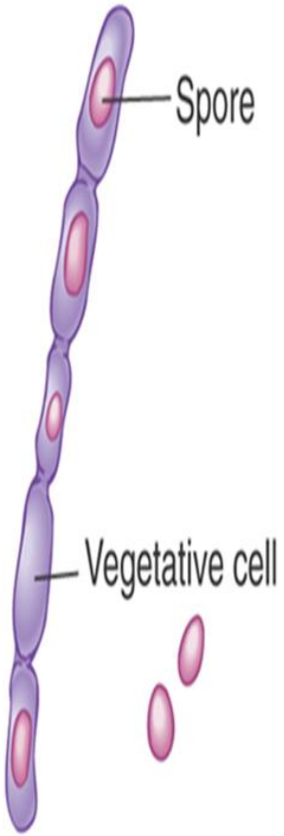
Important Species

1-Bacillus anthracis: The causative agent of Anthrax.

2-Bacillus cereus: A common cause of Food Poisoning

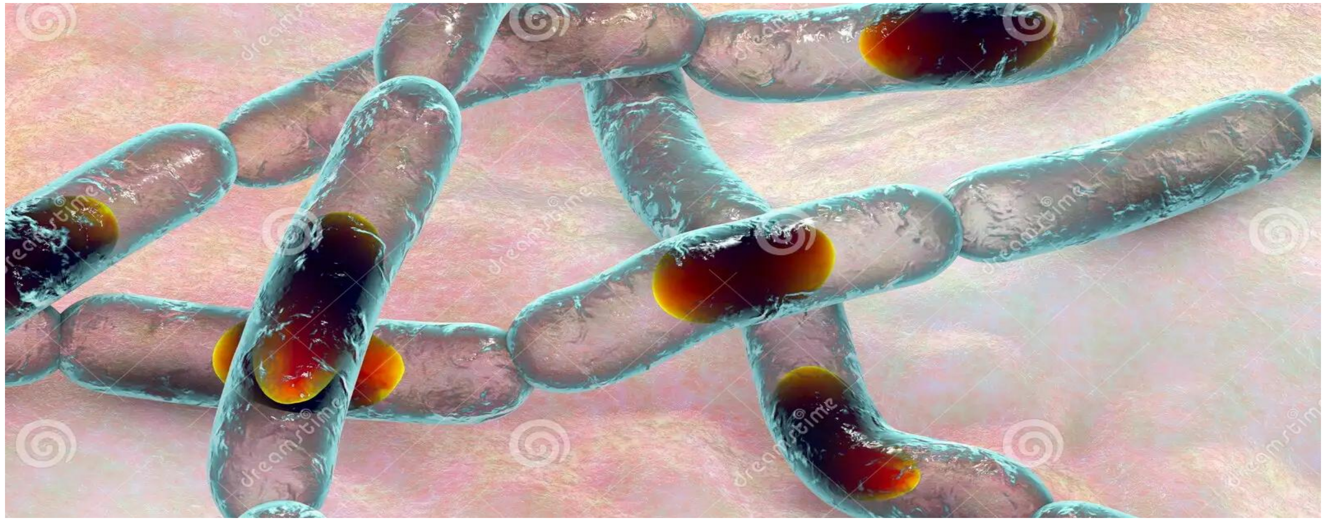
Spore Formation: This is their most important characteristic: they produce resilient Endospores. These spores are highly resistant to heat, desiccation, and disinfectants, making them difficult to kill.

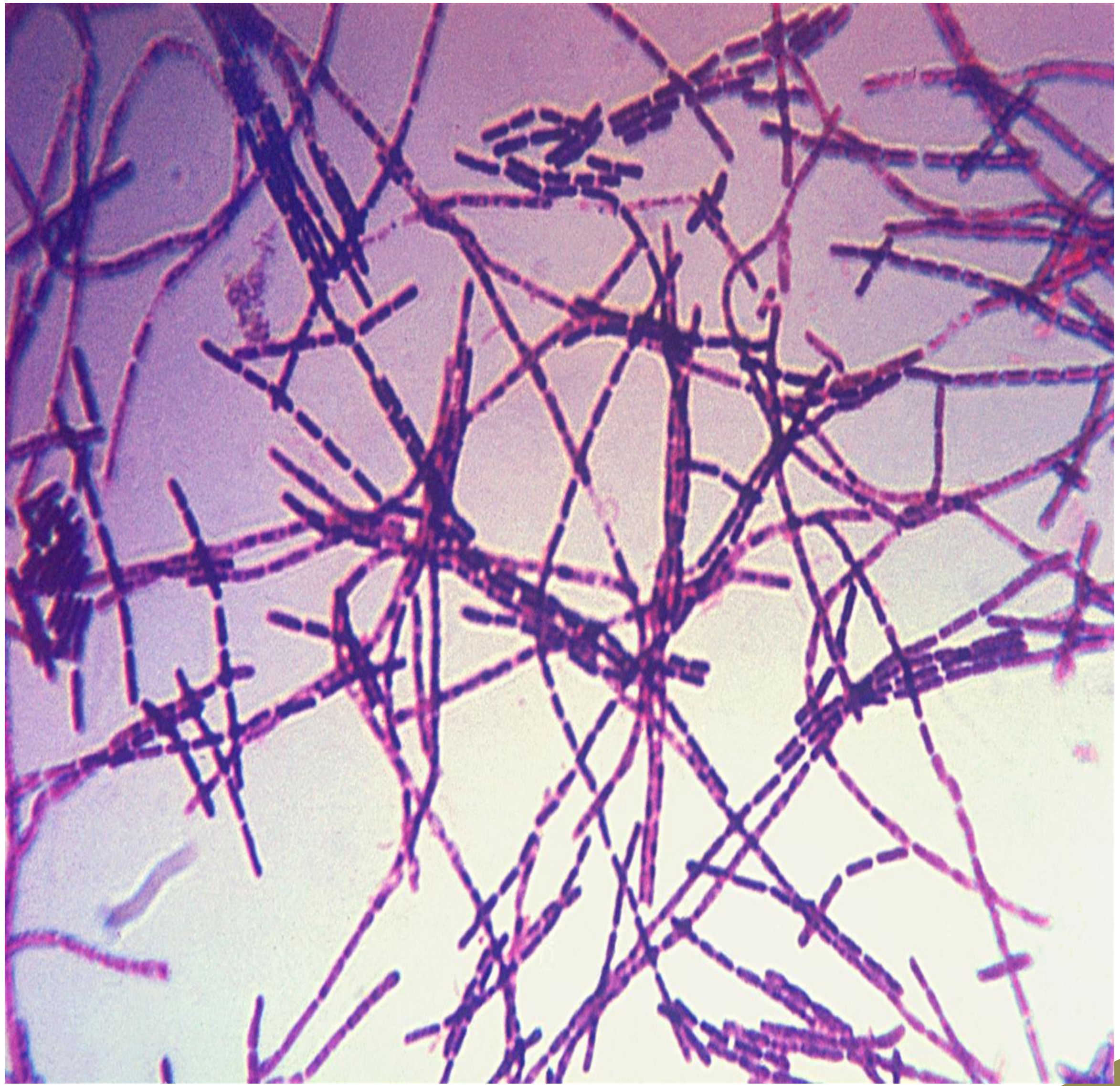
Non motile spore forming (the location of the spore is either central, terminal, or subterminal according to species).



Bacillus anthracis

1. **Morphology:** Large G+ve rods, arranged in **long chain**, with **square end** (bamboo stick) appearance.
2. **Spore-forming:** central spore, the same width of the cell.
3. **Non hemolytic**
4. **Capsulated** (polypeptide)





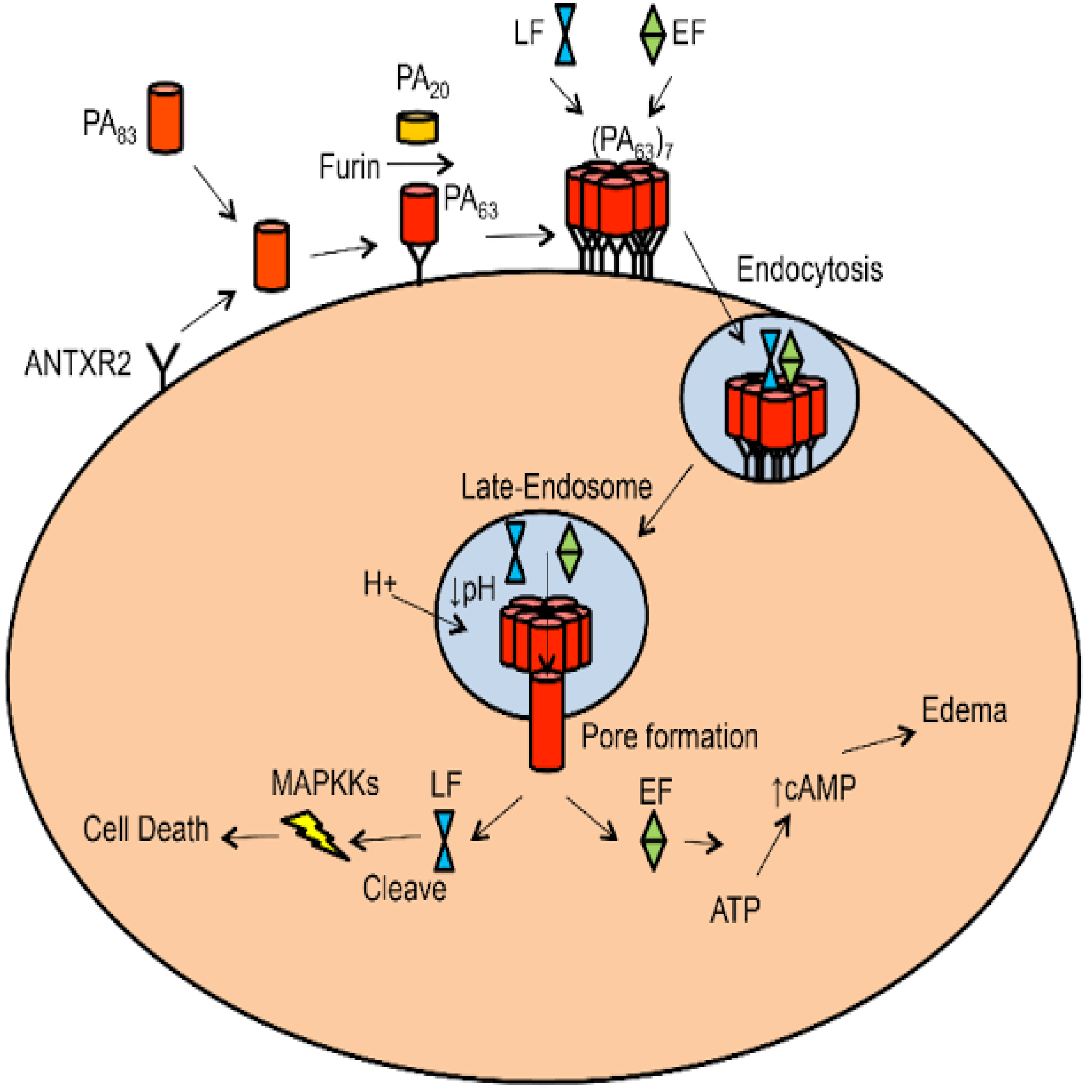
Virulence factors:-

1- Capsule: poly- D- glutamic acid capsule, **antiphagocytic.**

2- Anthrax toxin- complex: The anthrax toxin is a tripartite virulence complex composed of three distinct proteins that act synergistically to produce disease:

1-Protective Antigen (PA):

Functions as the binding and entry component of the toxin. It attaches to specific receptors on the host cell surface and forms a membrane pore, allowing the enzymatic factors (EF and LF) to enter the cytoplasm.



2-Edema Factor (EF):

An adenylate cyclase enzyme that increases intracellular cAMP levels, leading to marked cellular edema at the infection site. It also impairs neutrophil activity, reducing the host’s ability to control infection.

3-Lethal Factor (LF):

The primary virulence determinant of Bacillus anthracic.

It is a zinc-dependent protease that disrupts key signaling pathways, ultimately causing cellular death and contributing to fatal outcomes in infected host.

There are 3 Types of Anthrax:

1- Cutaneous (malignant pustule)– spores enter through skin, black sore-eschar; occur on arms or hands & less frequently on face & neck.

Stages of cutaneous anthrax:-

-Papule at the site of entry (wound or scratches).

-Vesicle

-Eschar: malignant pustule

(A)

(B)

2- Gastrointestinal anthrax:

-ingested spores, rare but commonly fatal disease.

- **Ulcers**
- **Bloody diarrhea**, due to necrosis and ulceration which produces GI hemorrhage.

-**Renal failure** due to anthrax toxin.

-**Death** in untreated cases (more than 50%)

Gastrointestinal anthrax

3- Pulmonary (wool sorter's disease)– inhalation of spores, most deadly form.

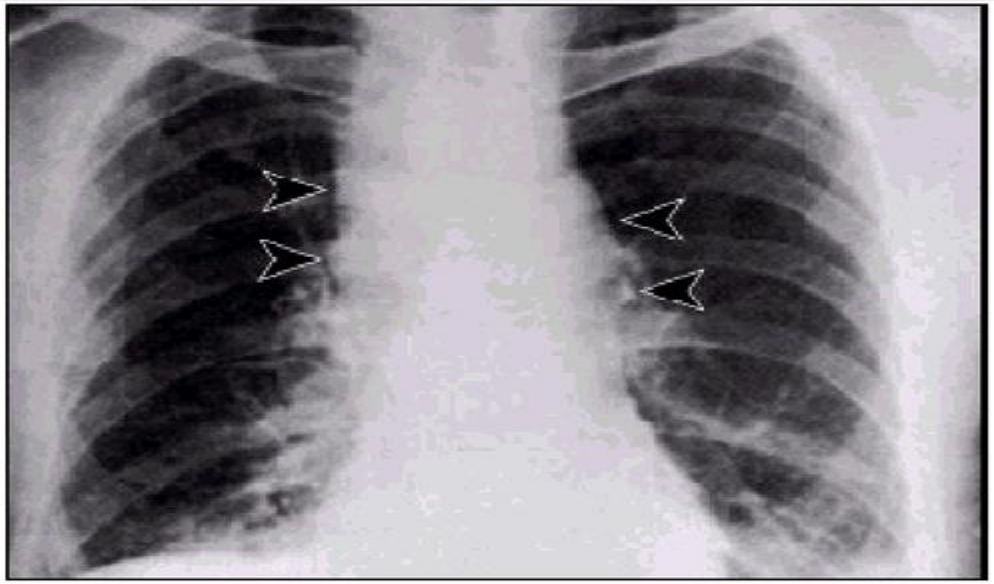
Inhalation anthrax:

symptoms: after 1-6 days incubation period **low grade fever** and **cough**.

Second phase: The early clinical manifestations of inhalation anthrax is marked after 1-2 days **hemorrhagic necrosis** & **edema** of the mediastinum & **substernal pain**, **high fever**, **shortness of breath**, **tachypnea**, and **hematemesis**, **Sepsis** may occur.

Spread to the meninges causing **hemorrhagic meningitis**.

The fatality rate is 85-90%.



Culture Characteristics

- On blood agar, *Bacillus anthracis* forms **non-hemolytic** colonies that are **large, raised, and gray-white** in appearance. The colony surface and irregular, filamentous margins give a characteristic **“Medusa head”** or **“beaten egg-white”** appearance.

Biochemical reaction:

- **glucose – maltose – sucrose fermentation**:-

B. anthracis produces acid without gases

Catalase	Positive (+ve)
Gram Staining	Positive (+ve)
Hemolysis	Negative (-ve)
Motility	Negative (-ve)
Gelatin Hydrolysis	Negative (–ve)

Ascoli's Test (Thermo precipitin Test):

A serological antigen–antibody precipitation test used to examine wool, hair, or tissues from decomposed carcasses for evidence of Bacillus anthracis. The test detects the polysaccharide antigen of B. anthracis in the sample.

Procedure:

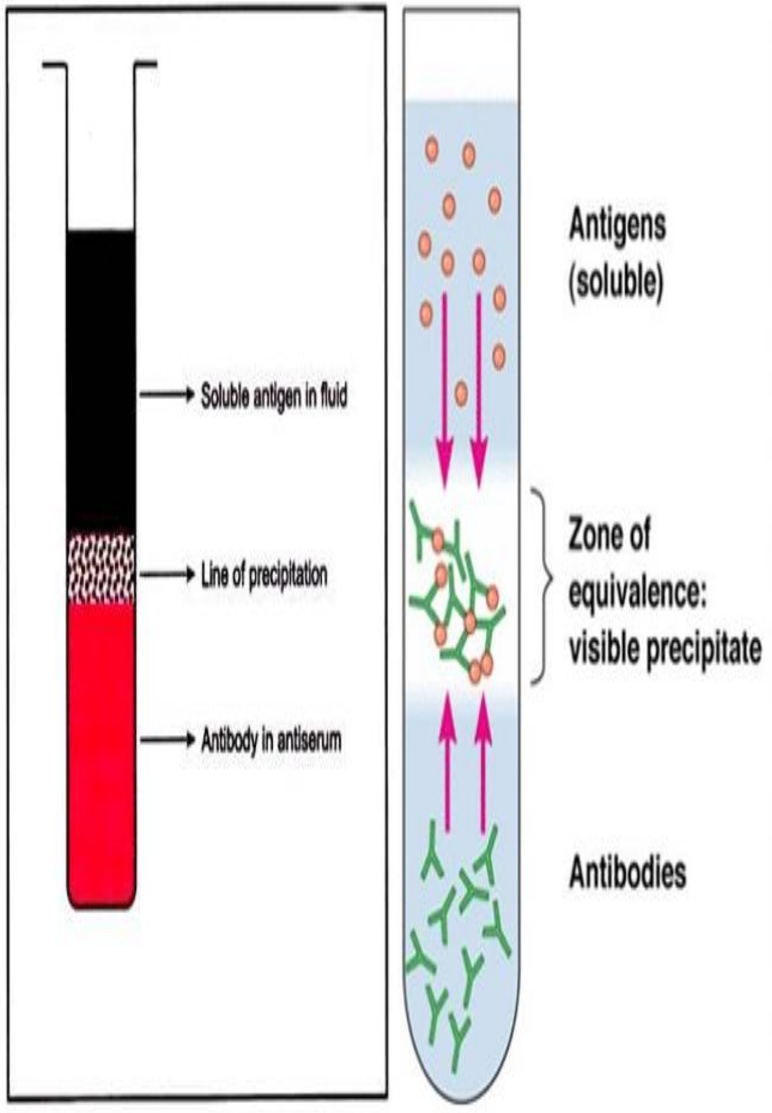
1-Take 2 cm of the suspected material and boil it for 10 minutes in 10 ml of sterile saline containing a few drops of 0.5% acetic acid.

2-Filter the mixture to obtain a clear extract.

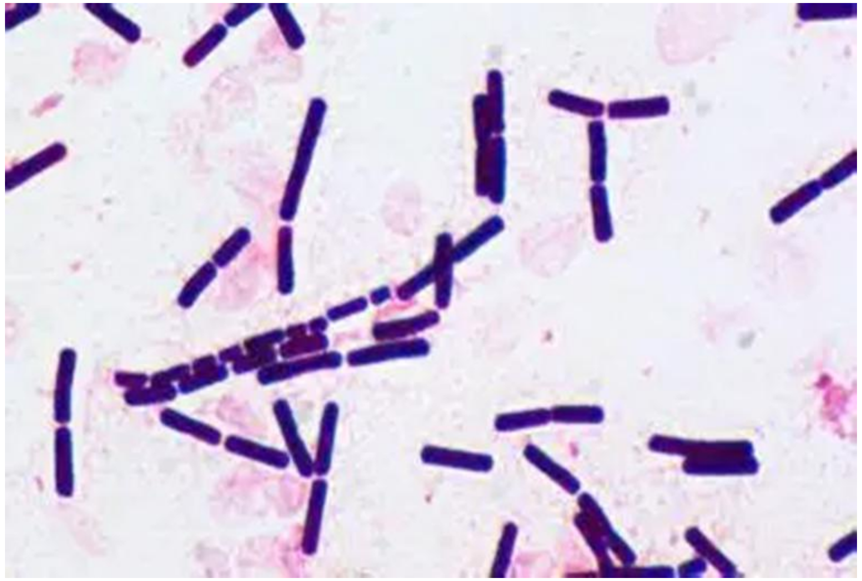
3-Place 0.5 ml of the filtrate into a narrow tube or capillary tube.

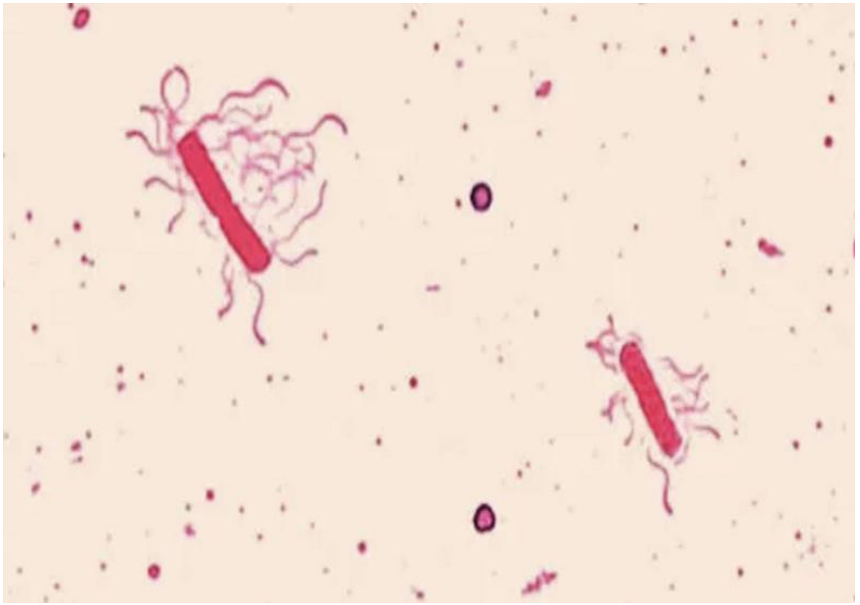
4-Carefully add 0.5 ml of anti-anthrax serum.

Positive Result: A distinct white precipitin ring forms at the interface between the filtrate and the antiserum.



Bacillus cereus





1-Gram-positive, rod-shaped morphology.

2-The cells typically have square ends.

3-Appear singly or in short chains, with clear junctions visible between the cells in chains .

-**spores :**oval, centrally to sub terminal located

-**motile** due to the presence of peritrichous flagella

-**Habitat:** Soil is the reservoir of this organism, It is also found in a variety of foods such as rice grains, potato, pasta, beans, and vegetables.

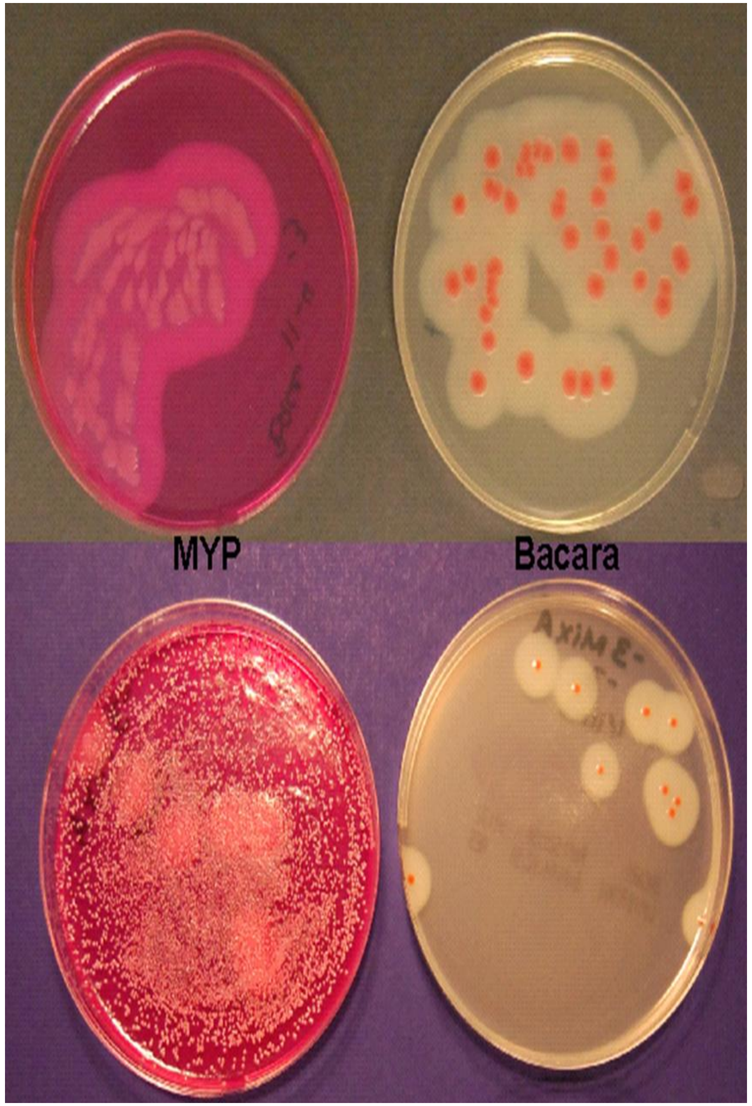
Cultural Characteristics:

MYP agar or **Bacara** agar is used to isolate *B. cereus* from food sample, which is both selective and differential media,

-After 18-24 hr. incubation at 30°C *B. cereus* colonies are usually a **pink-orange** color on Bacara or pink on MYP and may become more intense after additional incubation.

Biochemical reaction

Catalase	Positive (+ve)
Gram Staining	Positive (+ve)
Hemolysis	Positive (+ve)
Motility	Positive (+ve)
Indole	Negative (–ve)
Oxidase	Negative (–ve)
Gelatin Hydrolysis	Negative (–ve)



Pathogenesis of *Bacillus cereus* :

Bacillus species can cause **food poisoning**, which occurs in two distinct forms:

1. Diarrhea type:

Typically linked to contaminated meat dishes and sauces.

Incubation period: **1–24 hours**.

Presents with **profuse diarrhea**, **abdominal cramps and pain**, and may also cause **eye infections**.

2. Emetic type:

Commonly associated with **fried rice**.

Characterized by **nausea, vomiting, and abdominal cramps**.

Usually **self-limiting**, with recovery occurring within **24 hours**.

Any Question?

Thank you